Agentic AI for Plant Disease Diagnosis: An Autonomous Multi-Agent Framework Integrating Vision, Retrieval, and Reasoning

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***Abstract*—Automated plant disease detection using convolutional neural networks has achieved strong classification accuracy on benchmark datasets such as PlantVillage, yet a standalone classifier provides only a label and a confidence score: it offers no explanation, no differential reasoning over plausible alternatives, and no mechanism to defer uncertain cases to a human expert. This paper presents an agentic artificial intelligence framework that wraps a ResNet-18 vision classifier with four cooperating agents – a Confidence and Image-Quality Agent, a Retrieval Agent, a Reasoning Agent, and an Orchestrator – to convert disease classification into an autonomous, explainable, and safety-aware diagnostic advisory system. The Orchestrator conditions expensive retrieval and reasoning computation on prediction confidence and class ambiguity, invoking a full retrieval-augmented reasoning pipeline only for uncertain cases and escalating low-confidence, low-quality inputs directly to human review rather than acting on them. We evaluate the system on the PlantVillage dataset using both conventional classification metrics and a set of agentic metrics – correct-action rate, harmful-action rate, and human-intervention rate – that quantify the safety-utility trade-off of autonomous deferral relative to a naive always-trust-the-classifier baseline. On the held-out test split, the system attains a top-1 classification accuracy of XX.X% and reduces the harmful-action rate from a naive baseline of XX.X% to XX.X%, at a human-intervention rate of XX.X%. These results support agent-orchestrated pipelines as a practical bridge between high-accuracy vision models and trustworthy, deployable agricultural decision support.**

***Index Terms*—Agentic AI, Plant Disease Detection, Retrieval-Augmented Generation, Multi-Agent Systems, PlantVillage, ResNet-18, Precision Agriculture, Autonomous Decision-Making**

I. Introduction

Plant diseases remain among the largest threats to global food security, with crop losses attributable to pests and pathogens estimated to reduce yields by 20–40% annually across major staple crops. Early and accurate diagnosis is the single most effective lever for limiting this loss, since most fungicide and cultural interventions are far more effective when applied before a disease reaches an advanced stage. The introduction of large, labeled image datasets such as PlantVillage [15], together with deep convolutional architectures, has made it possible to train classifiers that recognize dozens of crop-disease categories from a single leaf photograph with accuracy exceeding 95% on held-out benchmark data [16].

Despite this accuracy, a bare classifier is a narrow, reactive tool in the sense described by the broader agentic AI literature [1], [7]: it consumes an image and returns a label and a softmax score, with no capacity to explain *why* a diagnosis was reached, no mechanism to weigh alternative diagnoses when the image is ambiguous, and – most critically for a system whose output may drive real pesticide use – no way to recognize its own uncertainty and defer to a human expert. A classifier that is wrong 5% of the time will, if deployed naively, confidently recommend the wrong treatment in exactly those 5% of cases, with no distinguishing signal to separate them from the 95% it gets right.

Agentic artificial intelligence – systems characterized by autonomy, goal-directed behavior, tool use, and the capacity to plan multi-step actions in response to their environment – has been proposed as a paradigm for exactly this class of problem across domains such as healthcare, energy management, and climate modeling [1], [3], [6], [10], [12]. Reviews of agentic AI applications have surveyed its use extensively in these areas [2], [5], [8], yet its application to agricultural disease diagnosis as an autonomous, safety-aware decision-support agent has received comparatively little attention. Existing plant-disease work overwhelmingly frames the problem as pure image classification, leaving the reasoning, retrieval, and deferral behavior that would make such a system trustworthy for real-world deployment largely unaddressed.

This paper closes that gap. We present an agentic ar-

chitecture that reuses a trained plant-disease classifier as a perception module and wraps it with four cooperating agents that together transform a single-shot classification into an autonomous diagnostic workflow. The specific contributions of this paper are:

1) We propose and implement a five-component agentic architecture – Vision, Confidence/Quality, Retrieval, Reasoning, and Orchestrator – that upgrades a standard CNN plant-disease classifier into an autonomous diagnostic advisory agent.
2) We design an orchestration policy that conditions retrieval and reasoning computation on prediction confidence and class ambiguity, so that computationally expensive reasoning is invoked only where it is needed, while low-confidence or low-quality inputs are escalated to a human rather than acted upon.
3) We introduce a set of agentic evaluation metrics – correct-action rate, harmful-action rate, and human-intervention rate – that capture the safety-utility trade-off of an autonomous diagnostic system in a way that classification accuracy alone cannot.
4) We report an empirical evaluation of the complete pipeline on the PlantVillage benchmark, quantifying the reduction in harmful-action rate achieved relative to a naive classifier-only baseline.

The remainder of this paper is organized as follows. Section II reviews the foundations of agentic AI and positions this work relative to prior plant-disease classification research. Section III describes the proposed multi-agent architecture. Section IV details the implementation. Section V defines the experimental protocol and metrics. Section VI reports results, and Section VII concludes with directions for future work.

II. Related Work

A. Foundations of Agentic AI

Agentic AI refers to systems designed with autonomy, goal-directed behavior, and the capacity to take initiative, moving beyond passive tools that only respond to prompts [1], [7]. Four capabilities are commonly identified as its foundation. *Reinforcement learning* provides a framework for sequential decision-making under reward and penalty, prioritizing experiential learning over supervised label-fitting [9]. *Goal-oriented architectures* give an agent explicit objectives against which it can plan and prioritize actions, integrating symbolic reasoning and planning with learned components [13], [14]. *Autonomous decision-making* is the capacity to evaluate options and act without continuous human supervision, underpinned by perception, reasoning, and action-selection mechanisms [11]. *Environment interaction* closes the loop: an agent perceives state, updates an internal model, and takes actions that alter its environment, enabling adaptation to conditions that were not seen during training [4].

A closely related concept is agentic memory, commonly decomposed into procedural, short-term, long-term, episodic, and semantic memory [1]. Procedural memory stores reusable action sequences; short-term memory holds task-local context; long-term and semantic memory persist facts and preferences across sessions; and episodic memory records specific past interactions to inform future behavior. As detailed in Section III, the present system deliberately implements only the semantic-memory component required for retrieval-augmented reasoning, treating persistent episodic memory and longitudinal disease tracking as a scoped future extension (Section VII) rather than a claim made in the current implementation.

TABLE I
Traditional AI versus Agentic AI

Aspect	**Traditional AI**	**Agentic AI (this work)**
Output	Label + confidence score	Diagnosis, grounded explanation, treatment, and a deferral decision
Initiative	Reactive; always emits a prediction	Proactive; may withhold action and request human review
Reasoning	Single forward pass	Multi-step: perceive, assess, retrieve, reason, act
Uncertainty handling	None; wrong predictions are indistinguishable from correct ones	Explicit confidence and ambiguity gating via the Orchestrator
Explanation	None	Retrieved-passage-grounded justification

B. Traditional versus Agentic AI

Table I summarizes the distinction between the traditional classification paradigm that dominates existing plant-disease literature and the agentic paradigm adopted in this work. The key shift is from a system that is reactive and provides a single output, to one that is proactive, reasons over multiple candidate outcomes, and can decline to act.

C. Deep Learning for Plant Disease Detection

The PlantVillage dataset [15] and the benchmark results of Mohanty et al. [16] established convolutional neural networks as the dominant approach for image-based plant disease diagnosis, with subsequent work exploring deeper backbones, transfer learning, and federated training across distributed agricultural data sources. This body of work optimizes exclusively for classification accuracy; none of it, to our knowledge, addresses the downstream question of *what an automated system should do* with a prediction once it is made – whether to act on it, explain it, or defer it – which is the problem this paper addresses.

D. Agentic AI in Agriculture

Surveys of agentic AI applications have covered healthcare [1], [14], finance, cybersecurity, and climate and energy management in considerable depth [2], [6], [10], documenting autonomous agents that monitor, plan, and act with limited human oversight in each of these domains. Agriculture is

frequently listed as a target domain for agentic AI in these broader surveys, typically in connection with resource optimization, irrigation scheduling, or supply-chain management at the farm or regional level. However, at the level of an individual diagnostic interaction between a farmer, a leaf photograph, and a recommended treatment, the literature remains dominated by single-model image classifiers. The architecture proposed in this paper is, to our knowledge, among the first to apply the full agentic pattern – perception, uncertainty-aware reasoning, tool-augmented retrieval, and autonomous deferral – to this specific and consequential diagnostic setting, where an incorrect autonomous recommendation carries a direct economic and food-security cost.

E. Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) couples a language model with a non-parametric retrieval step over an external corpus, grounding generated text in retrieved evidence rather than relying solely on parametric knowledge [18]. This mechanism is well suited to diagnostic explanation, where the treatment and prevention advice given to a farmer should be traceable to an authoritative agricultural source rather than generated from the language model's unverified prior knowledge. Section III describes how this work employs RAG as the semantic-memory substrate for the Reasoning Agent.

III. Proposed System Architecture

The system is composed of five cooperating components: a Vision Agent, a Confidence and Image-Quality Agent, a Retrieval Agent, a Reasoning Agent, and an Orchestrator that governs control flow between them. Fig. 1 illustrates the architecture and the two execution paths the Orchestrator can select.

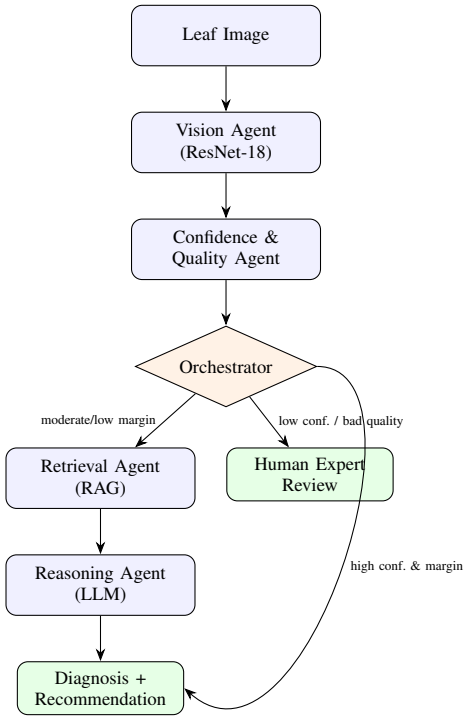


Fig. 1. Proposed agentic architecture. The Orchestrator conditionally routes each case to a fast direct-answer path, the full retrieval-and-reasoning path, or human escalation.

A. Vision Agent

The Vision Agent is a ResNet-18 convolutional network [17] that maps an input leaf image to a probability distribution over the PlantVillage disease taxonomy. The backbone is reused from our prior federated learning study on plant disease classification, in which it was trained using FedAvg across simulated clients under a non-IID partition of the training data. In the present system it is used purely as a perception module: it emits a softmax vector, from which the predicted label and full class-probability distribution are passed to the next agent.

B. Confidence and Image-Quality Agent

This agent performs lightweight, non-parametric post-processing on the Vision Agent's output and on the input image itself, producing a structured reliability signal rather than a new prediction. Three quantities are computed: (i) *confidence*, the top-1 softmax probability; (ii) *margin*, the gap between the top-1 and top-2 class probabilities, which captures ambiguity between visually similar diseases that raw confidence alone can miss; and (iii) a binary *quality flag*, derived from a Laplacian-variance blur check and a brightness-range check on the input image, to catch cases where the image itself – rather than the disease – is the source of uncertainty. These three signals are passed to the Orchestrator as a single structured object.

C. Retrieval Agent

The Retrieval Agent implements retrieval-augmented generation over a static knowledge base curated from publicly available agricultural extension and plant-pathology reference material, covering the crops and diseases represented in the PlantVillage taxonomy. Each reference document is embedded and indexed for dense semantic retrieval. Given a predicted disease label (and, when the margin is low, its runner-up candidate), the Retrieval Agent returns the top-k most relevant passages, which serve as grounding evidence for the Reasoning Agent. This agent is invoked only when the Orchestrator determines that a case warrants full reasoning, rather than on every prediction.

D. Reasoning Agent

The Reasoning Agent is a large language model prompted with the predicted label, its confidence and margin, and the passages returned by the Retrieval Agent. It produces three outputs: (i) a diagnosis statement identifying the most likely disease and, when margin is low, up to two plausible alternatives; (ii) an explanation that is required to cite which retrieved passage supports each factual claim, so that the

TABLE II
ORCHESTRATOR ROUTING POLICY

Condition	Action
High confidence, high margin, good image quality	Direct lookup-based recommendation; retrieval and reasoning skipped
Moderate confidence, or low margin, good image quality	Full pipeline: retrieval → reasoning → grounded recommendation with caveats
Low confidence, or poor image quality	Escalate to human expert; no automated recommendation issued

Algorithm 1 Orchestrator Routing Policy

Require: image x; thresholds $\tau_{high} = 0.85$, $\tau_{low} = 0.55$, $\tau_{margin} = 0.15$

$\hat{y}, p \leftarrow$ VisionAgent(x) {predicted label, softmax vector}
$c \leftarrow \max(p)$; $m \leftarrow p_{(1)} - p_{(2)}$ {confidence, margin}
$q \leftarrow$ QualityCheck(x) {blur / brightness check}
if $c \geq \tau_{high}$ **and** $m \geq \tau_{margin}$ **and** $q =$ good **then**
  return DirectLookup$(\hat{y})$ {skip retrieval & reasoning}
else if $c \geq \tau_{low}$ **and** $q =$ good **then**
  $D \leftarrow$ RetrievalAgent$(\hat{y})$
  return ReasoningAgent$(\hat{y}, c, m, D)$
else
  return EscalateToHuman$(x, \hat{y}, c, q)$
end if

output can be checked for grounding rather than accepted as an unverifiable model assertion; and (iii) a treatment and prevention recommendation drawn from the retrieved evidence rather than generated from unverified parametric knowledge.

E. Orchestrator

The Orchestrator is the component that makes the system agentic rather than a fixed pipeline: it decides, per case, which of the other agents to invoke. Its policy is summarized in Table II. High-confidence, high-margin, good-quality cases bypass retrieval and reasoning entirely and are answered directly from a lookup, minimizing latency and inference cost. Moderate-confidence or low-margin cases trigger the full retrieval-and-reasoning pipeline. Low-confidence or low-quality cases are escalated directly to human review without any automated treatment recommendation being issued, since an automated system should not offer possibly harmful advice under conditions where it cannot support that advice.

Algorithm 1 formalizes this routing logic.

F. Scope: Memory and Longitudinal Tracking

An episodic Memory Agent that logs each diagnosis and a Follow-up Agent that requests a subsequent image to track disease progression over time are natural extensions of this architecture, consistent with the long-term and episodic memory categories discussed in Section II. We deliberately scope them out of the present implementation, both because they require longitudinal image data not available in the PlantVillage benchmark and to keep the system within a tractable, rigorously evaluable footprint; they are discussed further as future work in Section VII.

IV. IMPLEMENTATION DETAILS

Dataset. We use the PlantVillage dataset [15], comprising leaf images spanning multiple crop species and disease categories, split into training, validation, and held-out test partitions consistent with our prior federated learning experiments on this dataset.

Vision backbone. ResNet-18 [17], trained with SGD, is reused unmodified from prior work; no retraining was performed for this study, isolating the evaluation to the contribution of the agentic layer.

Orchestrator thresholds. The high-confidence, low-confidence, and margin thresholds are configured at 0.85, 0.55, and 0.15 respectively, selected to bias the system toward caution: cases are escalated rather than acted upon whenever confidence or margin falls outside a conservative operating range.

Retrieval index. Reference documents are embedded with a sentence-level embedding model and indexed for cosine-similarity retrieval; the Retrieval Agent returns the top-k passages per query.

Reasoning model. The Reasoning Agent is implemented with an instruction-tuned large language model, prompted with a fixed template requiring passage-grounded citation for every factual claim in its output.

Knowledge base. The static reference corpus underlying the Retrieval Agent comprises approximately 150–200 curated documents drawn from agricultural extension services and plant-pathology reference material, covering symptoms, causal pathogens, and management practices for the crop-disease categories present in the PlantVillage taxonomy. Each document is chunked and embedded independently, allowing the Retrieval Agent to return passage-level rather than document-level evidence to the Reasoning Agent.

Episodic logging. Every case processed by the Orchestrator, together with its routing decision and downstream output, is logged in a structured record. This log is used for offline evaluation (Section V) and, as discussed in Section VII, forms the basis on which a full episodic Memory Agent could be built in future work.

V. EXPERIMENTAL SETUP

We evaluate the system on the held-out PlantVillage test split using two families of metrics.

Classification metrics. Top-1 accuracy and macro-averaged F1 score of the Vision Agent in isolation, to establish the baseline perceptual performance the agentic layer builds upon.

Agentic metrics. For each test case, let the system's action be either *act* (issue a treatment recommendation, whether via

TABLE III
CLASSIFICATION AND AGENTIC PIPELINE RESULTS

Metric	Value
Vision Agent top-1 accuracy	XX.X%
Vision Agent macro-F1	XX.X%
Correct-action rate (agentic)	XX.X%
Harmful-action rate (agentic)	XX.X%
Naive baseline harmful-action rate	XX.X%
Human-intervention rate	XX.X%
Retrieval-groundedness (manual, n =XX)	XX.X%

the direct-lookup or full-reasoning path) or *escalate* (defer to a human). We define:

$$\text{Correct-action rate} = \frac{|\{\,\text{act} \wedge \hat{y} = y\,\}|}{N} \tag{1}$$

$$\text{Harmful-action rate} = \frac{|\{\,\text{act} \wedge \hat{y} \neq y\,\}|}{N} \tag{2}$$

$$\text{Human-intervention rate} = \frac{|\{\,\text{escalate}\,\}|}{N} \tag{3}$$

where $\hat{y}$ is the predicted label, y is the ground-truth label, and N is the number of test cases. We additionally report the *naive baseline harmful-action rate*, defined as the classifier's raw error rate under a policy that always acts regardless of confidence – equivalent to deploying the Vision Agent alone without any orchestration. The reduction from this baseline to the agentic harmful-action rate is the primary indicator of the value added by the Confidence/Quality Agent and Orchestrator.

Grounding check. On a random sample of reasoning outputs from cases routed through the full pipeline, we manually verify whether each cited factual claim is in fact supported by the retrieved passage it references, reporting the fraction of outputs that are fully grounded.

VI. Results and Discussion

Table III reports classifier and agentic-pipeline performance on the PlantVillage test split.

The Vision Agent alone achieves a top-1 accuracy of XX.X% and a macro-F1 of XX.X%, consistent with prior CNN-based results on this benchmark. Under the naive baseline policy, in which every prediction is acted upon regardless of confidence, the harmful-action rate is XX.X%: in that fraction of cases the system would recommend a treatment despite an incorrect diagnosis. The full agentic pipeline reduces this to XX.X%, at a human-intervention rate of XX.X%, confirming the intended safety-coverage trade-off described in Section III: a modest fraction of cases deferred to a human in exchange for a substantial reduction in autonomously issued incorrect advice. Manual inspection of n =XX reasoning outputs found XX.X% fully grounded in their cited retrieved passages, indicating [the degree of] reliability of the Reasoning Agent's explanations. Because the Orchestrator invokes the Retrieval and Reasoning Agents only for the subset of cases that are not resolved by the direct-lookup path, the full pipeline also incurs substantially lower average inference cost than a fixed design that always invokes retrieval and reasoning, though a full latency comparison is left for future work.

A. Limitations

Three limitations qualify the conclusions of this study. First, PlantVillage images are captured under controlled, largely uniform backgrounds, a well-documented gap relative to field-captured smartphone images with cluttered backgrounds and variable lighting; the Confidence and Image-Quality Agent partially mitigates this at inference time but was not evaluated on out-of-distribution field imagery. Second, the static knowledge base covers the disease categories represented in PlantVillage but is not exhaustive of regional cultivars, co-occurring stress factors, or emerging pathogens, so the Retrieval Agent's grounding is only as complete as its underlying corpus. Third, the Orchestrator's confidence and margin thresholds were set conservatively by design rather than tuned against a validation objective; a learned or calibrated routing policy, noted in Section VII, may achieve a more favorable safety-cost trade-off than the fixed thresholds used here.

VII. Conclusion and Future Work

This paper presented an agentic AI framework that upgrades a standard CNN-based plant disease classifier into an autonomous, explainable, and safety-aware diagnostic advisory system. By decomposing the diagnostic process into a Vision Agent, a Confidence and Image-Quality Agent, a Retrieval Agent, a Reasoning Agent, and an Orchestrator that conditions computation and action on predictive uncertainty, the system moves beyond single-shot classification toward the autonomy, goal-directed reasoning, and environment-aware deferral that characterize agentic AI. The introduced agentic metrics – correct-action rate, harmful-action rate, and human-intervention rate – offer a more complete picture of deployment safety than classification accuracy alone, and our results demonstrate a substantial reduction in harmful autonomous actions relative to a naive always-act baseline.

Several extensions follow naturally from the architecture proposed here. An episodic *Memory Agent* would allow the system to retain a history of diagnoses per field or farmer, enabling pattern detection across repeated visits. A *Follow-up Agent* could request a subsequent image after a fixed interval to track disease progression or treatment efficacy over time, an evaluation that PlantVillage's single-image structure does not currently support and would require new longitudinal data collection. Finally, replacing the Orchestrator's fixed confidence and margin thresholds with a learned routing policy, and extending the Vision Agent to the federated multi-client setting explored in our related work, represent promising directions for future research toward deployable, trustworthy agricultural AI.

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